Robust Two-Stage Radiotherapy Treatment Planning

Jonathan Wolloch
Advisor: Dr. Shimrit Shtern

Faculty of Data and Decision Sciences
Technion - Israel Institute of Technology

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Outline

1. Handling Uncertain Data With Robust Optimization
2. Optimizing Radiotherapy Treatment Planning
3. Numerical Experiments
4. Future Directions

Handling Uncertain Data With Robust Optimization

1. We would like to assume that good data leads to good decisions.
2. However, as reported by Ben-Tal and Nemirovski 2000, for some LPs even a 0.01% perturbation in the data may cause certain constraints to become highly infeasible.

Several Key Observations

1. At the time of decision making, data isn't known exactly.
2. We cannot ignore this uncertainty.
3. A methodology for handling this uncertainty is needed.

The discussion herein, regarding robust optimization, relies on Ben-Tal, Ghaoui, and Nemirovski 2009.

Enter Robust Optimization (For the LP case)

A generic LP can be characterized with the triplet (A, b, c).

$$\min_{x \in \mathbb{R}^n} \{c^T x : Ax \leq b\}$$

In the robust optimization framework, an *uncertain-LP* is defined as a collection:

$$\min_{x \in \mathbb{R}^n} \{c^T x : Ax \leq b, \forall (A, b, c) \in \mathcal{U}\}$$

Under this paradigm, we do not persume to know the data's distribution, only the range of possible values.

What does it mean to solve an uncertain LP?

The Underlying Decision-Making Environment

1. x represents "here-and-now" decisions.
2. The decision-maker is fully responsible for the consequences of the decisions, only when the actual data is within $\mathcal{U}$.
3. The decision-maker cannot tolerate violations of the constraints when the data is in $\mathcal{U}$.

An "uncertainty immunized" solution

In this environment, an uncertainy immunized solution is a fixed vector that remains feasible whatever the realization of the data within $\mathcal{U}$.

We call such a solution *robust feasible*.

What is the value of a *robust feasible* solution?

Since we want a solution that is feasible no matter the realization, it is natural to attribute the worst-case value of the uncertain-LP to the robust feasible solution, that is:

$$\sup\{c^T x : (A, b, c) \in \mathcal{U}\}$$

Thus, the best *robust feasible* solution can be found by solving the following LP:

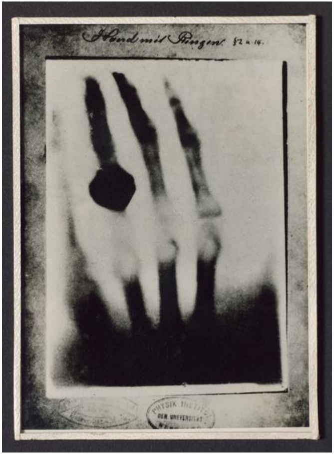
$$\min_x \{ \sup_{(A,b,c)\in\mathcal{U}} c^T x : Ax \leq b, \ \forall (A, b, c) \in \mathcal{U}\}$$

or equivalently:

$$\min_{x,t} \{t : c^T x \leq t, \ Ax \leq b, \ \forall (A, b, c) \in \mathcal{U}\}$$

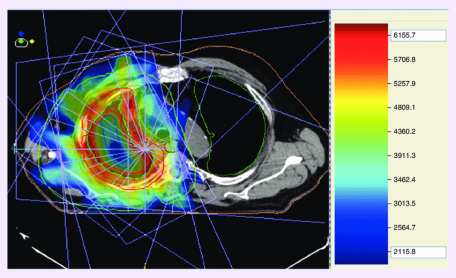
Discovery of X-rays

In 1895, Wilhelm Conrad Röntgen documented his discovery of X-rays, earning him the Nobel Prize in Physics in 1901.



Applications of X-rays

Röntgen's discovery laid the foundation for a variety of medical procedures, including both medical imaging and radiotherapy.



Planning a Radiotherapy Treatment Session

1. We aim to deliver a high radiation dose to the tumor.
2. We would like to deliver a low radiation dose to healthy tissues.
3. We want the radiation dose delivered to the tumor to be relatively uniform.

Where does the uncertainty come from?

The efficacy of radiotherapy is determined not only by the intensity of the radiation applied, but also by biological factors, most notably the oxygenation level (Hill et al. 2015).

The level of oxygenation is estimated using medical images, which are prone to measurement and implementation errors. It also varies with the patient's condition. In short, it is highly uncertain.

Previous Work

Goldberg, Langer, and Shtern 2024 applied the robust optimization framework to the problem of optimal radiotherapy treatment.

The authors assumed the scope of the treatment is a single session.

The uncertainty set used in this work bounded the difference in radiosensitivity between tumorous voxels by a function of their distance.

Optimizing a Radiotherapy Treatment Plan - Notation

1. $[m]$ - the set of all voxel indices.
2. $T \subseteq [m]$ - the set of all tumor voxels.
3. $H_k \subseteq [m]$ - the set of all voxels in the k-th healthy organ, $\forall\, k \in [K]$.
4. n - the number of beamlets whose intensity is to be determined.
5. $\mathbf{D} \in \mathbb{R}^{m \times n}$ - the dose matrix, where $\mathbf{D}_{i,j}$ is the influence the j-th beamlet has on the i-th voxel.
6. N - the number of fractions.
7. $\mathbf{x} \in \mathbb{R}^{N \times n}$ - the intensity of the n beamlets in each of the N fractions.

Note: $T \cup \bigcup_{k=1}^{K} H_k = [m]$

Modeling the Radiosensitivity

The so-called *physical dose* applied to a voxel v at fraction f is given by:

$$d_{f,v}(\mathbf{x}) = \sum_{i=1}^{n} \mathbf{D}_{v,i} x_{f,i}$$

1. $\mathbf{D}_{v,i}$ is the influence the i-th beamlet has on the voxel v.
2. The impact of the *physical dose* depends on biological factors.

This information is modeled via a *radiosensitivity vector* $\phi \in [0,1]^{N \times T}$. Accordingly, the *bio-adjusted dose* applied to a voxel v at fraction f is given by:

$$\phi_{f,v} \cdot d_{f,v}(\mathbf{x})$$

Optimizing a Radiotherapy Treatment Plan – Nominal Formulation

Taking $\hat{\phi}$ at face value, we can formulate the following LP:

$$
\begin{aligned}
\max_{\mathbf{x}\in\mathbb{R}_+^{N\times n},\ \underline{d}\in\mathbb{R},\ \underline{d}^F\in\mathbb{R}} \quad & \underline{d} + \lambda \underline{d}^F && && (1.a)\\
\text{subject to} \quad & \underline{d}^F \le \hat{\phi}_{fv} d_{fv}(\mathbf{x}), && v \in T,\ f \in [N] && (1.b)\\
& \hat{\phi}_{fv} d_{fv}(\mathbf{x}) \le \mu^F \underline{d}^F, && v \in T,\ f \in [N] && (1.c)\\
& \underline{d} \le \sum_{f\in[N]} \hat{\phi}_{fv} d_{fv}(\mathbf{x}), && v \in T && (1.d)\\
& \sum_{f\in[N]} \hat{\phi}_{fv} d_{fv}(\mathbf{x}) \le \mu \underline{d}, && v \in T && (1.e)\\
& d_{fv}(\mathbf{x}) \le \bar{d}_k^F, && f \in [N],\ k \in [K],\ v \in H_k && (1.f)\\
& \sum_{f\in[2]} d_{fv}(\mathbf{x}) \le \bar{d}_k, && k \in [K],\ v \in H_k && (1.g)
\end{aligned}
$$

Optimizing a Radiotherapy Treatment Plan – Uncertainty Set

Here we apply the robust optimization methodology by assuming that the values of ϕ are members of the following uncertainty set, henceforth referred to as the *spatially and temporally bound uncertainty set*:

$$\mathcal{U}_{\text{STB}} = \left\{ \phi \in [0,1]^{N \times |T|} \;\middle|\; \begin{array}{ll} |\phi_{1v} - \hat{\phi}_{1v}| \leq \delta, & \forall v \in T \\ |\phi_{fv} - \phi_{fu}| \leq \gamma_{vu}, & \forall f \in [N],\ v, u \in T \\ |\phi_{fv} - \phi_{(f+1)v}| \leq \sigma, & \forall f \in [N-1],\ v \in T \end{array} \right\}$$

- δ - the maximum allowed deviation from the measured radiosensitivity vector.
- γ_{vu} - parameters given by a metric-preserving function of the distance between v and u. Limits the maximum allowed deviation between the radiosensitivity values of two consecutive fractions.
- σ - the maximum allowed deviation between the radiosensitivity value of a voxel across two consecutive fractions.

Optimizing a Radiotherapy Treatment Plan – Robust Formulation

After a couple of simple algebraic manipulations, we can write the *robust counterpart* of the nominal LP:

$$\max_{\mathbf{x}\in\mathbb{R}_+^{N\times n},\ \underline{d}\in\mathbb{R},\ \underline{d}^F\in\mathbb{R}} \quad \underline{d} + \lambda \underline{d}^F \tag{2.a}$$

$$\text{subject to} \quad \min_{\phi\in\mathcal{U}_{\mathrm{STB}}} \{\phi_{fv} d_{fv}(\mathbf{x})\} \geq \underline{d}^F, \qquad v \in T,\ f \in [N] \tag{2.b}$$

$$\max_{\phi\in\mathcal{U}_{\mathrm{STB}}} \{\phi_{fv} d_{fv}(\mathbf{x}) - \mu \phi_{fu} d_{fu}(\mathbf{x})\} \leq 0, \qquad v, u \in T,\ f \in [N] \tag{2.c}$$

$$\min_{\phi\in\mathcal{U}_{\mathrm{STB}}} \left\{ \sum_{f\in[N]} \phi_{fv} d_{fv}(\mathbf{x}) \right\} \geq \underline{d}, \qquad v \in T \tag{2.d}$$

$$\max_{\phi\in\mathcal{U}_{\mathrm{STB}}} \left\{ \sum_{f\in[N]} \phi_{fv} d_{fv}(\mathbf{x}) - \mu \sum_{f\in[N]} \phi_{fu} d_{fu}(\mathbf{x}) \right\} \leq 0, \qquad v, u \in T \tag{2.e}$$

$$d_{fv}(\mathbf{x}) \leq \bar{d}_k^F, \qquad f \in [N],\ k \in [K],\ v \in H_k \tag{2.f}$$

$$\sum_{f\in[2]} d_{fv}(\mathbf{x}) \leq \bar{d}_k, \qquad k \in [K],\ v \in H_k \tag{2.g}$$

Optimizing a Radiotherapy Treatment Plan – Reformulation

The robust counterpart is a semi-infinite program — that is, a program with an infinite number of constraints — and must be reformulated to be solvable.

Lemma 1

Let $\mathcal{U} \subseteq \mathbb{R}^n$ be a Reformulated Robust set. Then for any continuous function $f : \mathbb{R}^{|\mathcal{I}|} \to \mathbb{R}$,

$$\min_{\mathbf{u}_{\mathcal{I}} \in \mathcal{U}} \{f(\mathbf{u}_{\mathcal{I}})\} = \min_{\mathbf{u}_{\mathcal{I}} \in \mathcal{P}_{\mathcal{I}}(S)} \{f(\mathbf{u}_{\mathcal{I}})\}$$

Lemma 2

The projection of $\mathcal{U}_{STB}$ onto ϕ_{fv} is

$$P_{fv}(\mathcal{U}_{STB}) = \left[\min_{\phi_{fv} \in \mathcal{U}_{STB}} \{\phi_{fv}\}, \max_{\phi_{fv} \in \mathcal{U}_{STB}} \{\phi_{fv}\} \right] \quad \forall\, v \in T,\ f \in [2]$$

Optimizing a Radiotherapy Treatment Plan – 1-D Projections

Lemma 3

Consider the case where $N = 2$*, and suppose for any* $u, v \in T$*, we have* $\gamma_{uv} = \Gamma(dist(u, v))$*, where* $dist(u, v)$ *is the voxel-to-voxel distance, and* $\Gamma : \mathbb{R} \to [0, 1]$ *is a nondecreasing, subadditive function satisfying* $\Gamma(x) = 0 \iff x = 0$*. Define for all* $v \in T$*:*

$$\underline{\phi}^0_{1v} = \max\left\{0,\ \hat{\phi}_{1v} - \delta\right\}, \qquad \bar{\phi}^0_{1v} = \min\left\{\hat{\phi}_{1v} + \delta,\ 1\right\} \tag{1}$$

$$\underline{\phi}_{1v} = \max_{u\in T}\left\{\underline{\phi}^0_{1u} - \gamma_{uv}\right\}, \qquad \bar{\phi}_{1v} = \min_{u\in T}\left\{\bar{\phi}^0_{1u} + \gamma_{uv}\right\} \tag{2}$$

$$\underline{\phi}_{2v} = \max\left\{0,\ \underline{\phi}_{1v} - \sigma\right\}, \qquad \bar{\phi}_{2v} = \min\left\{\bar{\phi}_{1v} + \sigma,\ 1\right\} \tag{3}$$

Then,

$$\max_{\phi\in\mathcal{U}_{STB}} \phi_{fv} = \bar{\phi}_{fv}, \qquad \min_{\phi\in\mathcal{U}_{STB}} \phi_{fv} = \underline{\phi}_{fv}$$

Optimizing a Radiotherapy Treatment Plan - Reformulated Constraint 2.b

Corollary 1

Constraint 2.b can be rewritten as:

$$\underline{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) \geq \underline{d}^F, \quad v \in T,\ f \in [2] \tag{3.b}$$

Optimizing a Radiotherapy Treatment Plan - Reformulated Constraint 2.d

Corollary 2

Constraint 2.d can be rewritten as:

$$\sum_{f \in [N]} \underline{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) \geq \underline{d} \tag{3.d}$$

Optimizing a Radiotherapy Treatment Plan – 2-D first fraction

Lemma 4

Suppose $v \in T$, and let $\underline{\phi} \in \mathbb{R}^{2\times|T|}$, $\bar{\phi} \in \mathbb{R}^{2\times|T|}$ have components given by (2) and (3), and suppose that for any $u, v \in T$,

$$\gamma_{uv} = \Gamma\big(dist(u, v)\big),$$

where $dist(u, v)$ is the voxel-wise distance, and $\Gamma : \mathbb{R} \to [0, 1]$ is a nondecreasing, subadditive function satisfying $\Gamma(x) = 0 \iff x = 0$.
Then, the projection of $\mathcal{U}_{STB}$ onto (ϕ_{1v}, ϕ_{1u}) satisfies:

$$\mathcal{P}_{1v,1u}(\mathcal{U}_{STB}) = \mathcal{U}^{(1)}_{v,u} \equiv \left\{ \ (\phi_{1v}, \phi_{1u}) \in \mathbb{R}^2 \ : \ \underline{\phi}_{1v} \leq \phi_{1v} \leq \bar{\phi}_{1v}, \ \underline{\phi}_{1u} \leq \phi_{1u} \leq \bar{\phi}_{1u}, \ |\phi_{1v} - \phi_{1u}| \leq \gamma_{vu} \ \right\}$$

Optimizing a Radiotherapy Treatment Plan – 2-D second fraction

Lemma 5

Suppose $v \in T$, and let $\underline{\phi} \in \mathbb{R}^{2\times|T|}$, $\bar{\phi} \in \mathbb{R}^{2\times|T|}$ have components given by (2) and (3). Suppose also that for any $u, v \in T$,

$$\gamma_{uv} = \Gamma\big(dist(u,v)\big),$$

where $dist(u,v)$ is the distance between voxels u and v, and $\Gamma : \mathbb{R} \to [0,1]$ is a nondecreasing, subadditive function satisfying $\Gamma(x) = 0 \iff x = 0$.
Then, the projection of $\mathcal{U}_{STB}$ onto (ϕ_{2v}, ϕ_{2u}) is:

$$\mathcal{P}_{2v,2u}(\mathcal{U}_{STB}) = \mathcal{U}^{(2)}_{v,u} \equiv \left\{ \ (\phi_{2v}, \phi_{2u}) \in \mathbb{R}^2 \ : \ \underline{\phi}_{2v} \le \phi_{2v} \le \bar{\phi}_{2v}, \ \underline{\phi}_{2u} \le \phi_{2u} \le \bar{\phi}_{2u}, \ |\phi_{2v} - \phi_{2u}| \le \gamma_{vu} \ \right\}$$

Optimizing a Radiotherapy Treatment Plan - Reformulated Constraint 2.c

Observation 1

Invoking proposition 7 from Apte et al. 2024, we can replace constraint 2.c by the constraint pair:

$$\bar{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) + \max\left\{\underline{\phi}_{fu},\ \bar{\phi}_{fv} - \gamma_{uv}\right\} \cdot (-\mu^F d_{fu}(\mathbf{x})) \leq 0, \quad v, u \in T,\ f \in [N] \tag{2.c$_1$}$$

$$\max\left\{\underline{\phi}_{fu} + \gamma_{uv},\ \bar{\phi}_{fv}\right\} \cdot d_{fv}(\mathbf{x}) + \underline{\phi}_{fu} \cdot (-\mu^F d_{fu}(\mathbf{x})) \leq 0, \quad v, u \in T,\ f \in [N] \tag{2.c$_2$}$$

Reformulated Robust Optimization Problem (Without commulutive homogeneity)

$$\max_{\mathbf{x}\in\mathbb{R}_+^{2\times n},\ \underline{d}\in\mathbb{R},\ \underline{d}^F\in\mathbb{R}} \quad \underline{d} + \lambda \underline{d}^F \tag{3.a}$$

subject to

$$\underline{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) \geq \underline{d}^F, \qquad v \in T,\ f \in [2] \tag{3.b}$$

$$\bar{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) + \max\left\{\underline{\phi}_{fu},\ \bar{\phi}_{fv} - \gamma_{uv}\right\} \cdot (-\mu^F d_{fu}(\mathbf{x})) \leq 0, \qquad v, u \in T,\ f \in [2] \tag{3.c'}$$

$$\max\left\{\underline{\phi}_{fu} + \gamma_{uv},\ \bar{\phi}_{fv}\right\} \cdot d_{fv}(\mathbf{x}) + \underline{\phi}_{fu} \cdot (-\mu^F d_{fu}(\mathbf{x})) \leq 0, \qquad v, u \in T,\ f \in [2] \tag{3.c''}$$

$$\sum_{f\in[2]} \underline{\phi}_{fv} \cdot d_{fv}(\mathbf{x}) \geq \underline{d}, \qquad v \in T \tag{3.d}$$

$$d_{fv}(\mathbf{x}) \leq \bar{d}_k^F, \qquad f \in [2],\ k \in [K],\ v \in H_k \tag{3.e}$$

$$\sum_{f\in[2]} d_{fv}(\mathbf{x}) \leq \bar{d}_k, \qquad k \in [K],\ v \in H_k \tag{3.f}$$

Reformulated Robust Optimization Problem

Numerical Experiments

Goldberg, Langer, and Shtern 2024 tested their framework on real data from two patients' brains, as well as synthetic liver data.

In this project, the numerical experiments were tested on the same liver data. The number of beamlets was 458.

Dataset	**PTV**	**OAR**			
	$\lvert T \rvert$	k	Name	$\lvert H_k \rvert$	$\bar{d}_k$ bounds
Liver	6,954	1	Liver	77,657	60
		2	Heart	32,983	50
		3	Entrance	146,462	60

How Can We Solve This Huge LP?

Naively generating all constraints results in a problem with 918 variables and over 194 million constraints (194,224,632 to be exact).

To avoid building an enormous LP, we use a row-generation algorithm: We first solve the LP without any homogeneity constraints. Next, we iteratively do the following:

1. Evaluate the neglected homogeneity constraints
2. Identify the most violated ones
3. Add them to the model
4. Re-solve the updated LP

Until all homogeneity constraints are satisfied

Computational Details

1. The code was implemented in Python 3.11.
2. The models were built and solved using a highly optimized implementation of the Primal-Dual Simplex algorithm, supplied by the commercial solver Gurobi (version 12.0.2).
3. The experiments were run on a machine equipped with an Intel Xeon Gold 6354 CPU (36 cores, 72 threads).

Radiosensitivity vs. Voxel Distance (Brain)

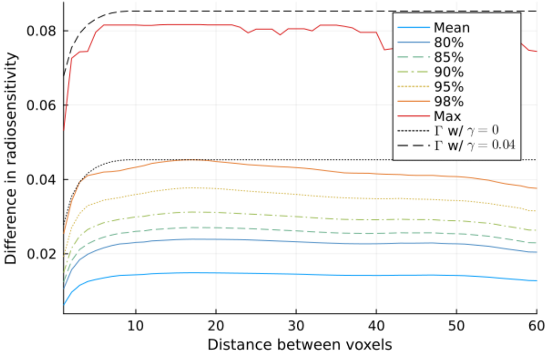


Figure: Radiosensitivity difference vs. voxel distance for PTV voxel pairs of Patient 4 in the TCIA archive Brain dataset.

Liver data was synthetically generated based on this data.

Computational Results Comparison

Metric	**10 constraints***	**5 constraints***
Solver time	4509.69 sec	4997.02 sec
Peak memory usage	8232.82 MB	8289.08 MB
Minimum fractional dose ($\underline{d}^F$)	11.864776	11.864776
Minimum total dose ($\underline{d}$)	23.042053	23.042053
Feasible solution found	Yes	Yes
Total constraints added	312,966	188,108

$$\mu^F = 1.3; \ \lambda = 0.5$$

***Maximum number of constraints added per voxel per iteration.**

Constraints Added per Iteration

Iteration	Run 1 – 3C1	Run 1 – 3C2	Run 2 – 3C1	Run 2 – 3C2
1	88,350	97,912	44,197	52,064
2	5,974	43,014	4,900	24,401
3	1,853	36,675	1,592	16,809
4	986	16,614	948	12,109
5	864	13,433	651	10,755
6	340	5,256	690	7,031
7	41	1,424	462	3,971
8	57	173	274	3,664
9	0	0	286	1,965
10	–	–	48	487
11	–	–	34	310
12	–	–	187	128
13	–	–	100	0
14	–	–	45	0
15	–	–	0	0

Future Directions

- Aggregate multiple voxels and refine locally.
- Think about how our decisions at fraction 1 will change given that we know that another scan will be received ahead of fraction 2.
- Incorporating total homogeneity constraints (possible approach using network flow).
- Generalizing to n fractions.